Student Performance Prediction and Learning Pattern Analysis

This project analyzes student academic performance to identify learning patterns and predict whether a student will pass or fail the final exam.

First, the Student Performance Dataset was collected from Kaggle and loaded into Python using Pandas. Basic data exploration and cleaning were performed to understand the dataset structure, check for missing values, and examine the distribution of exam scores.

Next, Exploratory Data Analysis (EDA) was conducted using visualizations such as histograms, boxplots, and correlation heatmaps. This helped understand relationships between variables like math score, reading score, and writing score. An average score was calculated for each student, and a pass/fail label was created based on whether the average score was above or below a threshold.

To analyze learning patterns, an unsupervised learning technique (K-Means clustering) was applied. Students were grouped into clusters based on their academic scores. These clusters represent different learning profiles such as consistent learners, average learners, and struggling learners.

After identifying student groups, supervised machine learning models were trained to predict whether a student will pass or fail. Two models were used:

- Logistic Regression
- Random Forest Classifier

The dataset was split into training and testing sets to evaluate the model performance. The models predicted student outcomes based on features such as exam scores and other factors.

To ensure reliable performance, cross-validation was used, which evaluates the model across multiple data splits. The models were evaluated using metrics such as accuracy, classification report, and confusion matrix.

The results show that machine learning techniques can effectively analyze student learning behavior and predict exam outcomes. This type of analysis can help educational institutions identify struggling students early and improve academic support strategies.

Output Interpretation (Proper Explanation)

The results show the performance of two machine learning models—Logistic Regression and Random Forest—used to predict whether a student will pass or fail the final exam.

Both models achieved an accuracy of 1.0 (100%), meaning that the models correctly classified all the students in the test dataset. Out of 200 students, every student's pass or fail outcome was predicted correctly.

The classification report provides additional evaluation metrics:

- Precision measures how many of the predicted results were correct.
- Recall measures how many of the actual cases were correctly identified.
- F1-score is the harmonic mean of precision and recall and indicates the balance between them.
- Support represents the number of actual samples in each class.

For Class 0 (Fail), the model correctly identified all 27 students who failed. The precision, recall, and F1-score for this class are all 1.00, indicating perfect predictions with no errors.

For Class 1 (Pass), the model correctly identified all 173 students who passed. Again, the precision, recall, and F1-score are 1.00, showing that every passing student was correctly classified.

The macro average and weighted average scores are also 1.00, which indicates that the model performed perfectly across both classes without any imbalance issues affecting the results.

Similarly, the Random Forest model also achieved 100% accuracy with identical precision, recall, and F1-score values. This indicates that both models were able to learn the relationship between the features and the target variable very effectively.

Overall, the results demonstrateh that the models were highly effective in predicting student performance in the dataset.